

DAT475 / DIT930 Advanced databases

2025-2026, Study Period 4

Assignment 3

Task description

In this assignment you will:

- Take data from a relational database (stored in a set of eleven CSV files that are given on the Assignment 1 page in Canvas) and recast it as RDF triples that conform to the OWL/RDFS that you produced in Assignment 2.
- Load the data into GraphDB.
- Query the data using SPARQL.

The aim of the first step listed above is to produce RDF triples that have class names, predicate names, and value types that match with the OWL/RDFS definition that you created in Assignment 2. How you create these files is up to you. One approach is just to write programs that take the CSV files as input and write out RDF files in a suitable format (e.g., turtle). I'm aware that some students have experimented with Ontotext Refine or Tarql in previous years; I don't know how successful those tools have been for this task.

Submitting work

You should submit:

- A PDF report that:
 - Describes how you approached this task.
 - Describes any decisions or assumptions that you made when producing RDF triples.
 - Gives SPARQL queries that answer the questions that are given on the Assignment 3 page in Canvas.
 - Gives the results obtained when those queries are executed.
- Any programs or scripts that you produced to generate data for loading into GraphDB.

In each file that you submit, give the names of the people submitting the work and your assignment group number (as given in Canvas).

Deadline: Tuesday 12 May 2026 at 23:59.